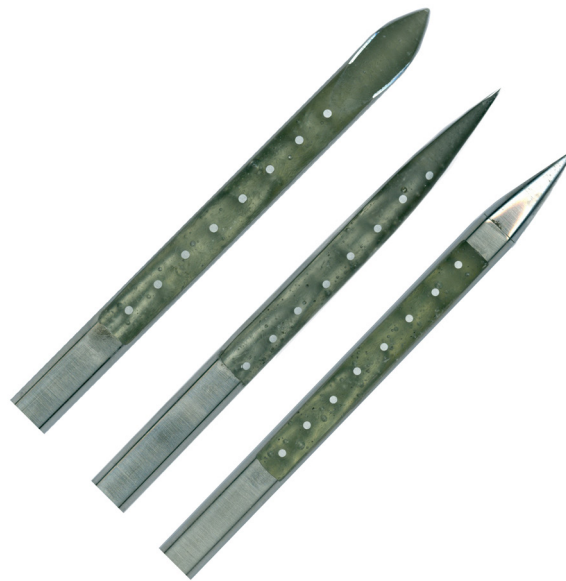


Probe Technical Guide

8, 16, 24, 32, and 64 Channels



Probe Technical Guide

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Documentation History

Date	Version	Notes
May 2017	PRBTN0003a	- Documented Created - Combined Probe Technical guides - Added S-probe - Removed PlexDrive - Updated Cleaning procedures and Jumpering information
January 2018	PRBTN0003b	-Reference Jumper instructions updated
December 2018	PRBTN0003c	-Updated text
September 2019	PRBTN0003d	-Updated for 64 channel probes
October 2020	PRBTN0003e	-Updated 64 channel probe information
August 2021	PRBTN0003f	-Updated stereotrode/tetrode schematic

Introduction

Plexon offers a suite of customizable specialty probes for various research needs. Flexibility in the manufacturing process allows for different electrode site arrangements and spacing, as well as optional fluid delivery channels for precise drug delivery or fiber optic lines for optogenetic stimulation intermixed within the recording sites. Please contact a Plexon Sales Engineer to discuss designing a customized probe.

Plexon probes are known for their robustness due in part to their stainless steel construction (as opposed to silicon), and ability to penetrate deep within the brain. When used according to proper procedures, probes will support many penetrations over a long period of time.

The U-Probe is a multi-use, multi-site linear electrode primarily used for acute studies with larger animals such as primates.

The V-Probe is a multi-use, multi-site, linear electrode. Compared with our U-Probe, the cone-shaped tip is believed to result in less trauma to the brain tissue upon penetration by minimally displacing tissue along the track. The design also permits an electrode site to be placed closer to the tip of the probe.

The S-Probe is a multi-use, multi-site, linear electrode with a solid stainless steel tip.

The Thumbtack Probe is a chronic, linear probe designed to span cortical layers running parallel to the surface of the brain.

Electrode Configurations

The U, V, and S Probes are available with single electrode configurations. In addition, the U-probe and S-Probe are available with stereotrode and tetrode configurations. Each configuration is available in 8, 16, 24, 32 and 64* channel counts. Each probe also features platinum/iridium electrode sites, and the option to be fitted with fluid capillaries (for fluid delivery) or optic fibers (for optical stimulation) as featured later in this document.

Single: The single electrode configuration is the most popular configuration and features the electrode sites in linear formation a



U-Probe/S-Probe Stereotrode: The stereotrode configuration describes the grouping of two electrode sites side-by-side.



U-Probe/S-Probe Tetrode: The tetrode configuration describes a grouping of four electrode sites. This configuration is most commonly used in areas with an especially high density of neurons (e.g. hippocampus).



*64 Channel neural probes are available in single and stereotrode configuration. 64 Channel probes with a single electrode configuration are available with 50, 75, or 100um inter-electrode spacing. 64 channel probes with stereotrode configuration require 100um or 150um interelectrode spacing.

Stimulation Options

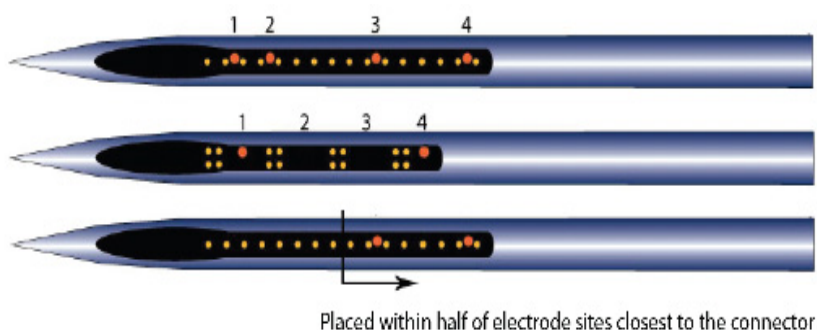
Up to four fluid capillaries and/or optic fibers may be added to a probe with single electrode configuration, and up to two fluid capillaries or optic fibers for U-Probe and S-Probes with stereotrode or tetrode configurations*. Placement of the fluid capillaries and/or optic fibers is largely up to the discretion of the researcher, within a few limitations. Fluid capillaries and/or optic fibers can be placed anywhere in the U-Probe or S-Probe, but may only be placed within the first half of channels on V-Probes. It is important to note that with the addition of each fluid capillary or optic fiber, the minimum diameter of the probe will likely need to increase to accommodate the additional materials (see the chart in the Technical Specifications section).

The illustrations below depict examples placements of fluid capillaries and/or optic fibers on probes with 16 channels.

4 fluid/optic channels,
single configuration U/S Probe

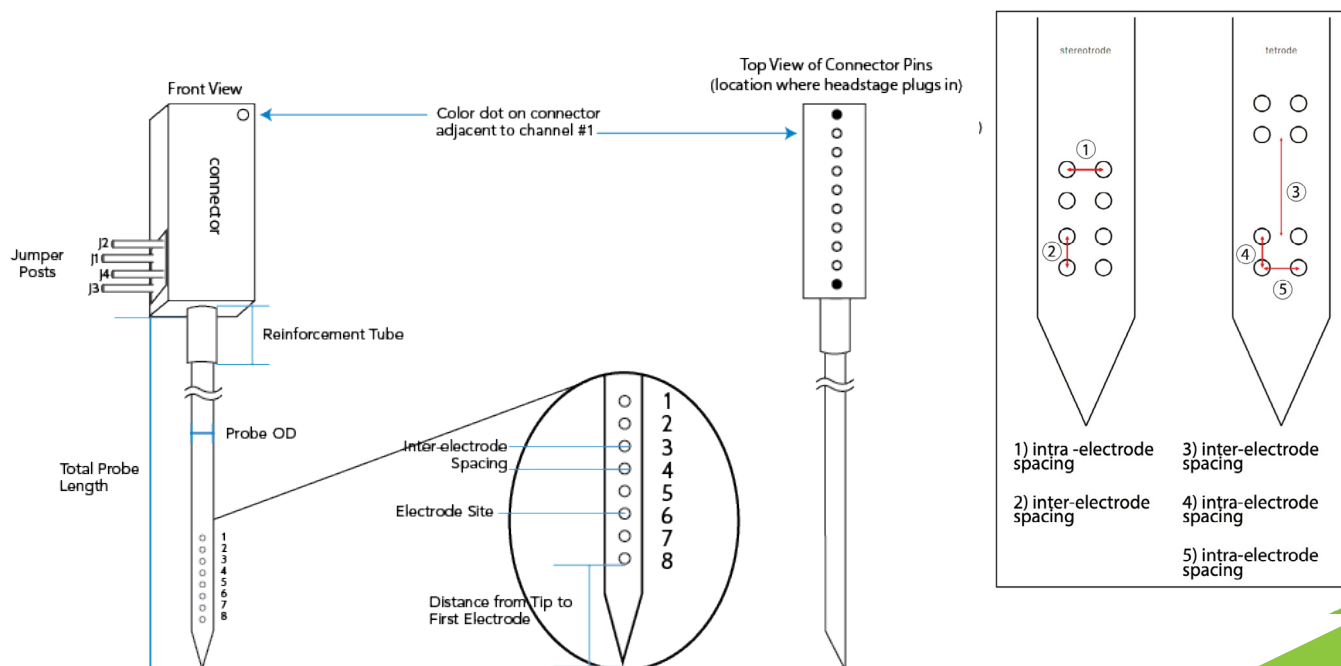
2 fluid/optic channels,
tetraode configuration U/S Probe

2 fluid/optic channel,
single configuration V-Probe



Probe Technical Specifications

The illustrations below depict the anatomy of a standard U-Probe by front and top (connector) view. The 8 channel, single electrode configuration is featured as the example. Electrode sites are numbered from connector to tip.



*64 channel single and stereotrode configurations can have one fluid capillary or one optic fiber

Probe Technical Specifications

Features	U-Probe	V-Probe	S-Probe	Remarks
Application	Acute			
Channel counts	8, 16, 24, 32, or 64			
Total probe length	30 - 150mm			Length is customizable within range provided; however, the most typical configuration is 100mm. Total probe length is measured from tip to the connector. The difference between the total probe length and the reinforcement tube length must be a minimum of 25 mm.
Probe diameters	185 - 500µm			Probe diameters vary based on the number of electrodes, fluid channels and optic fibers. Refer to Tables
Reinforcement tube length	5 - 125mm			Length is customizable within range provided
Reinforcement tube outer diameters	460 or 640µm			Refer to Table
Electrode construction	15, 20, 25 and 40µm diameter platinum/iridium electrode site			15µm diameter sites are recommended for resolving single units.
Electrode configurations	Single, Stereotrode, or Tetrode*	Single	Single, Stereotrode, or Tetrode*	Single configuration is most typical; stereotrodes and tetrodes tend to be used in areas with an especially high density of neurons (e.g. hippocampus). 64 channel available in stereotrode configuration with 100µm or 150µm interelectrode spacing.
Inter-electrode spacing	50, 75, 100, 150, or 200µm along length of probe			100, 150 and 200µm are most typical. 64 Channel probes with a single electrode configuration are available with 50, 75, or 100µm inter-electrode spacing. 64 channel probes with stereotrode configuration require 100µm or 150µm interelectrode spacing.
Distance from tip to the closest electrode site	Refer to Table	300µm	Refer to Table	The numbering of the electrode sites is such that channel #1 is closest to the connector. The minimum distance is a factor of tip angle and probe diameter.
Angled Reinforcement Tube	Yes			Enables the placement of multiple probes within close proximity of each other at the same time Angle can be 0-45° and will be placed in the middle of the reinforcement tube.
Fluid capillary ID, OD	40µm, 60µm			Fluid capillaries are slightly offset from the center line of the electrodes.
Optic fiber ID, OD, NA	46µm, 50µm, .66NA			Optic fibers are slightly offset from the center line of the electrodes.
Maximum number of fluid capillaries and/or optic fibers	Single: Up to 4 Stereotrode or Tetrode: Up to 2 64 channel: 1	4	Single: Up to 4 Stereotrode or Tetrode: Up to 2 64 Channel: 1	Single Electrode Configuration: Able to accommodate any combination of up to four additional fluid capillaries/optic fibers. U-Probe and S-Probe Stereotrode or Tetrode: Able to accommodate any combination of up to two additional fluid capillaries/optic fibers. 64 channel single and stereotrode configurations can have one fluid capillary or one optic fiber.
Connector interfaces	8 channel: CON/8o50m-10P 16 channel: 2x CON/8o50m-10P, CON/16m-V or CON/16o25m-18P 24 channel: 3x CON/8o50 or CON/32m-V 32 channel: CON/32m-V 64 channels: 2x CON/32m-V or CON/64f-Samtec			
Lifespan	Robust and reusable with a minimum of 30 penetrations, likely many more			Extensive, repeat use is expected upon using proper penetration procedures

*64 channel not available in tetrode configuration

Probe Technical Specifications

Single Electrode Configuration

Channels	Fluid Capillaries or Optic Fibers	Minimum Probe Diameter (μm)*	Distance from tip to 1st electrode**			Reinforcement Tube Diameter (μm)
			U-Probe	S-Probe	V-Probe	
8	0,1	185	320	500	300	460 or 640
8	2	210	360	540	300	460 or 640
8	3	260	450	640	300	640
8	4	300	520	710	300	640
16	0	210	320	500	300	460 or 640
16	1	210	360	540	300	460 or 640
16	2	236	410	580	300	640
16	3	300	520	710	300	640
16	4	320	560	750	300	640
24	0	210	360	540	300	460 or 640
24	1	236	410	580	300	640
24	2	300	520	710	300	640
24	3	320	560	750	300	640
24	4	360	620	820	300	640
32	0	260	450	640	300	640
32	1	260	450	640	300	640
32	2	300	520	710	300	640
32	3	320	560	750	300	640
32	4	360	620	820	300	640
64	0	320	560	750	300	640
64	1	360	620	820	300	640

Stereotrode/Tetrode Electrode Configuration

Channels	Fluid Capillaries or Optic Fibers	Minimum Probe Diameter (μm)	Distance from tip to 1st electrode*		Reinforcement Tube Diameter (μm)
			U-Probe	S-Probe	
8	0-1	185	320	500	460 or 640
8	2	210	360	540	460 or 640
16	0	210	320	500	460 or 640
16	1, 2	236	410	600	640
24	0	210	360	540	460 or 640
24	1, 2	260	450	640	640
32	0	260	450	640	640
32	1, 2	300	520	710	640
64***	0	320	560	750	640
64***	1	360	620	820	640

* Applicable to probes with 15μm diameter electrode sites

**Batch variation in tubing inner diameter may require a larger probe diameter or larger reinforcement tube

***Available in Stereotrode configuration with interelectrode spacing of 100μm or 150μm

Ground and Reference Jumper Instructions

Most Plexon probes have three or four configuration pins or sockets on the back of the connector block, and are supplied with “jumpers” that can be placed on those pins or into those sockets. Probes configured with the CON/8o50m-10P connector (typically 8 channel and 16 channel probes) will have four pins. Probes configured with CON/16m-V, CON/16o25m-18P, CON/32m-V, or CON/32o25m-36P connectors (24 channel and 32 channel) will have three or four sockets. The purpose of this section is to outline and illustrate the different probe grounding and referencing options using the pins or sockets.

Probes configured with CON/8o50m-10P (8 channel, 16 channel, or 24 channel)

In this example, there are four pins on the back-side of the Omnetics connector.

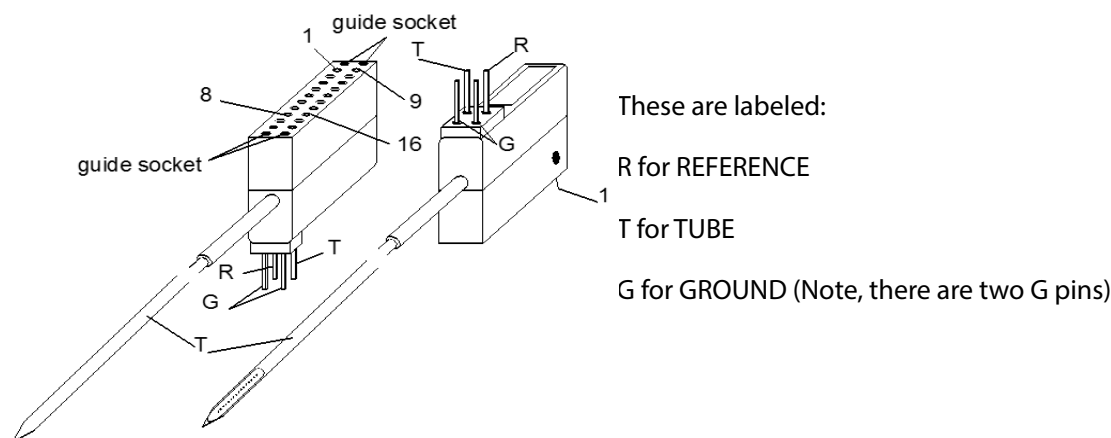
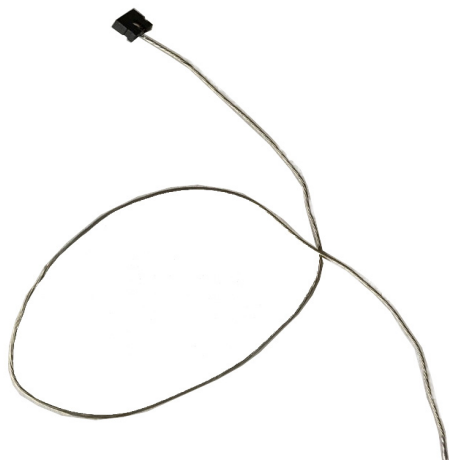


Illustration of 16 channel probe configured with two CON/8o50m-10P connectors; four pins

The TUBE pin is electrically connected to the stainless-steel shaft of the probe. The remaining three pins are not connected to the probe, instead they are connected directly to specific pins in the Omnetics connector, which interfaces to the recording headstage.

The pin marked REFERENCE is connected to the pin in the Omnetics connector that mates with the reference input of the headstage. The probe does not contain a dedicated reference electrode site.

Both of the pins marked GROUND are connected to each other and connected to the Omnetics connector that mates with the ground pin(s) of the headstage. These should be considered a source of “ground”, provided by the headstage, that can be used to ground other things around the electrode/animal. This is the 0V reference point of the headstage against which all other inputs to the headstage are measured.



Jumper included with probes configured with CON/8o50m-10P connectors

Practical Examples

Tube grounded, Reference grounded

When two jumpers are positioned vertically on the probe pins as shown in the picture at the right, one jumper connects the TUBE pin to one of the GROUND pins. This causes the TUBE of the probe to be grounded. The other jumper connects the REFERENCE pin to one of the GROUND pins. This causes the reference input to the headstage to be grounded.

If either jumper has a wire connected to it, that wire is grounded because both jumpers are touching a ground pin. Thus, the wire can serve as a source of “ground” to ground other metal objects near the subject. For example, it can be used to ground a guide tube through which the probe is inserted.



Tube used as the Reference electrode

When one jumper is positioned connecting the top two pins, as shown in the picture to the right, it connects the TUBE to REFERENCE. The reference input of the headstage is connected to the TUBE of the probe and the TUBE of the probe acts as a reference electrode.

If there is a wire connected to the top jumper, the wire is not ground because the jumper does not connect to either ground pin. Instead, the wire may act like an antenna and pick up stray signals that could pollute the reference. Do not use the jumper with a wire in this position.



If a second jumper is positioned horizontally connecting the bottom two pins as shown in the picture to the right, then it connects the two ground pins to each other. This second jumper serves no functional purpose. However, if there is a wire connected to this lower jumper, then that wire is grounded because the jumper is touching both ground pins. The wire may be used to “ground” other metal objects near the subject.

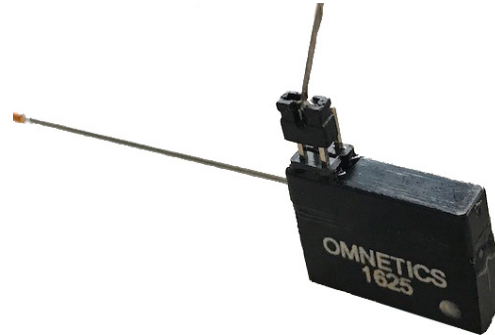


Tube Floating, Reference grounded

If one jumper is positioned across the pins as shown in the photo to the right, it connects the REFERENCE pin to one of the GROUND pins. The reference input to the headstage is grounded.

Since the TUBE pin is not connected to anything, the TUBE of the probe is not connected to anything (other than the tissue it is inserted into) and it is "floating".

Note that in many cases, the probe is passed through a guide tube. If the guide tube is grounded, physical contact between the probe and the guide tube may cause the tube of the probe to become grounded.



Probes configured with CON/16o25m-18P, CON/16m-V, CON/32m-V (16 channel, 24 channel, or 32 channel)

In this example, there are three sockets on the back-side of the Omnetics connector.

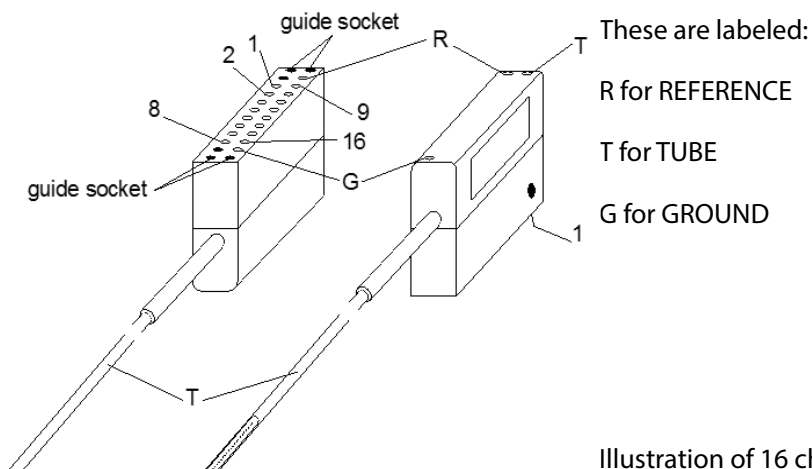


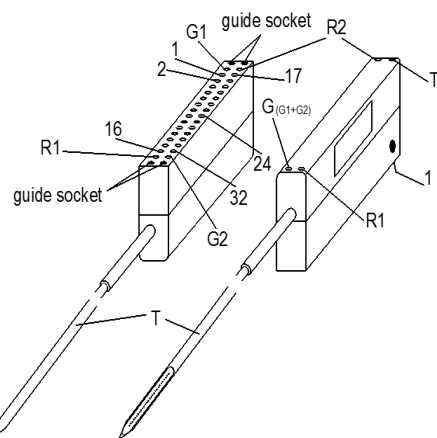
Illustration of 16 channel probe configured with one CON/16m-V connectors; three sockets

The TUBE socket is electrically connected to the stainless-steel shaft of the probe. The remaining sockets are not connected to the probe, instead they are connected directly to specific pins in the Omnetics connector, which interfaces to the recording headstage.

The socket marked REFERENCE is connected to the pin in the Omnetics connector that mates with the reference input of the headstage. The probe does not contain a dedicated reference electrode site.

The socket marked GROUND is connected to the pin in the Omnetics connector that mates with the ground pin(s) of the headstage. This should be considered a source of "ground", provided by the headstage, that can be used to ground other things around the electrode/animal. This is the 0V reference point of the headstage against which all other inputs to the headstage are measured.

Most 16 channel, 24 channel, and 32 channel probes have the three socket configuration described above. Some 24 channel and 32 channel probes may have a four socket configuration, as illustrated below.

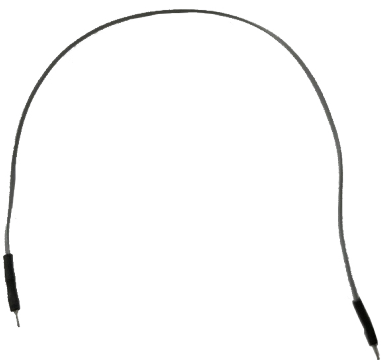


In this configuration, there are two different pins in the Omnetics connector that corresponds with two different reference inputs are required for the same headstage and recording. This sort of recording situation is not common.

different pin in the Omnetics connector. This configuration should only be used when two different reference inputs are required for the same headstage and recording. This sort of recording situation is not common.

Jumper Wires

Jumper wires included with probes configured with CON/16o25m-18P, CON/16m-V, and CON/32m-V connectors



Two-point jumper wire
(male – male)



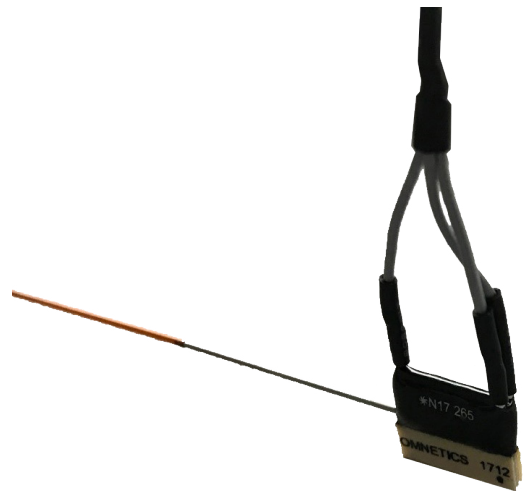
Four-point jumper wire
(male – male – male – female)

Practical Examples

Tube grounded, Reference grounded

When the four-point jumper is plugged into all three of the sockets, as shown in the photo to the right, the TUBE, REFERENCE, and GROUND are tied to the pin in the Omnetics connector corresponding to the GROUND. This will ground the tube of the electrode and reference of the headstage.

The probe shipment should also include a two-point jumper (male – male). This jumper wire can be attached to the female jumper on the four-point jumper wire. This wire would then serve as a source of “ground” to ground other metal objects near the subject. For example, it can be used to ground a guide tube through which the probe is inserted.



Tube used as the Reference electrode

Two points of the four-point jumper wire can be used to connect the TUBE and REFERENCE sockets. In this configuration, the reference input of the headstage is connected to the tube of the probe, and the tube of the probe acts as a reference electrode.

In this configuration, the jumper does not connect to the GROUND jumper socket; therefore, the wire is not a source of ground.



Tube floating, Reference grounded

Two points of the four-point jumper wire can be used to connect the REFERENCE and GROUND sockets. In this configuration, the reference input to the headstage is grounded.

Since the TUBE socket is not connected to anything, the TUBE of the probe is not connected to anything (other than the tissue it is inserted into), it is “floating”.

Note that in many cases, the probe is passed through a guide tube. If the guide tube is grounded, physical contact between the probe and guide tube may cause the tube of the probe to become grounded.



Dura Penetration

We do not recommend using probes to penetrate dura. If your experiments require that the probe penetrate dura, the S-probe would have the strongest tip.

Probe Cleaning Methods

After completion of the experiment and removal of the probe, it is critical that any proteins, lipids, blood, or other biological matter be removed in a timely manner. If the probe is not immediately cleaned, these substances may dry on the end of the probe, covering the sites. This dried layer will serve as an insulator and greatly increase the impedance in future experiments, resulting in decreased signal quality. It is also critical that the electrode sites are not physically wiped with any material such as a cloth or cotton swabs. Otherwise, it may have a detrimental effect on impedance.

Lastly, probes may be rinsed briefly with alcohol, but must be allowed to air dry immediately thereafter. The tips of the probe must not be allowed to sit in alcohol for more than 15 seconds as exposure to alcohol runs the risk of dissolving or damaging the epoxy which holds the electrode sites in place.

There are two methods to effectively clean probes: using enzymatic detergents (recommended) or Nolvasan disinfectant. One method is sufficient. There is no need to perform both.

Enzymatic Detergents

Plexon recommends using either Metrex EmPower Dual-Enzymatic Detergent (available at www.metrex.com) or Tergazyme from Alconox to clean probes after each use. These detergents contain protease enzymes which remove a variety of proteins, and surfactants to remove carbohydrates, lipids and other proteins. When used properly, the detergent will not damage the electrode sites. Please consult the manufacturer's webpage for more information.

NOTE: Be sure to follow all safety precautions on the container before proceeding.

Step 1: Dilute the detergent with warm water in the appropriate ratio (1:128 for Metrex) and swirl to mix. The enzyme activity is affected by temperature, so only water between 68 - 104°F (20 - 40°C) should be used. A fresh detergent solution should be made for cleaning after each experiment.

Step 2: Stir probe in the diluted detergent for approximately 30 seconds and allow it to soak for 30 minutes.

Step 3: Remove probe and stir it in distilled water to remove any detergent residue that might remain.

Step 4: Allow to air dry and place the probe in its container.

If it is not possible to immediately clean the probes in the recommended enzymatic detergents after use, the probe should be submerged in water until it can be thoroughly cleaned.

Nolvasan Disinfectant

Nolvasan disinfectant is another method of sterilization to consider.

Step 1: Place probe shaft and tip into diluted Nolvasan disinfectant solution for one hour.

Step 2: Run purified, distilled water over the probe.

Step 3: Allow to air dry and place the probe in its container.

Fluid Capillary Loading and Cleaning Procedures

It is important when utilizing a probe with a capillary to test the capillary before the first use. When you first receive a new probe and after each use, please wash the capillary with distilled water and blow air through in order to dry it and to avoid any remaining fluid residue. Lastly, never leave alcohol in the capillary for more than several minutes as this runs the risk of dissolving or damaging the epoxy.

Please note that the outer diameter of the reinforced portion of the fluidic tube is 0.36mm. The 0.28mm ID Polyethylene tubing, when stretched, will fit snugly over top of the 0.36mm OD probe fluidic inlet tube and 28 gauge hypodermic needle.

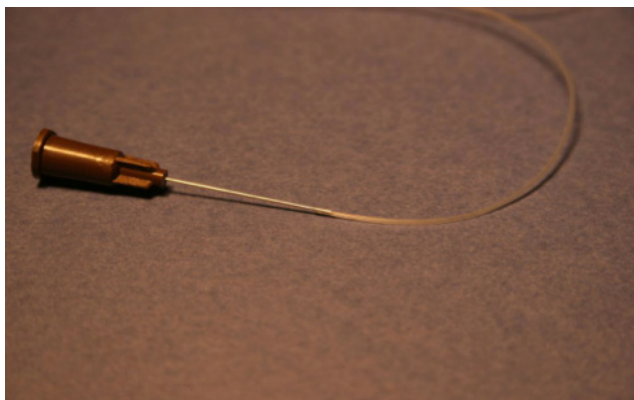
Materials:

- ♦ Three 28 gauge hypodermic needles (one each for drug injection, air cleaning and fluid cleaning).
- ♦ Three 10 - 30cm PORTEX Polythene inlet tubes with 0.28mm inner diameter (one for use with each of the three 28 gauge hypodermic needles).
- ♦ One 5µL Hamilton syringe for drug delivery (Type 75, N. tip 2).
- ♦ One Hamilton DS Digital Dispenser or homemade dispenser that fits the Hamilton syringe.
- ♦ One 1mL syringe for filling.
- ♦ One 5 - 10mL syringe for cleaning.
- ♦ Distilled and filtered water.
- ♦ Cotton tip applicator.

Loading Instructions (Steps 1 - 5):

Step 1: First prepare three sets of inlet tubes (one for the drug delivery, one for air cleaning and one for fluid cleaning) by attaching one inlet tube to each of the hypodermic needles (be sure you do not puncture the tube and it does not leak at the junction) — see picture below. The fluid going into the probe has to be distilled water or isopropyl alcohol based solution without solids (i.e. water soluble or isopropyl-soluble).

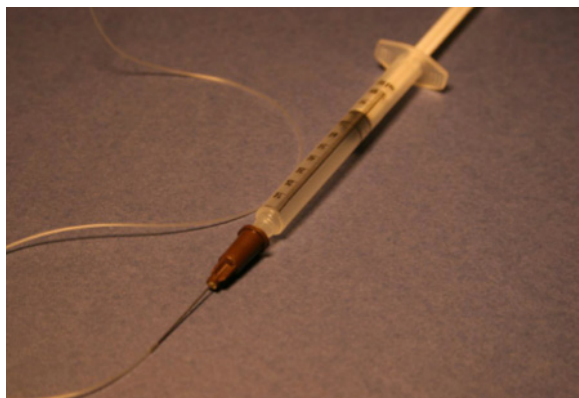
Step 2: Using one of the three prepared inlet tubes and the 1mL syringe, fill the syringe with drug



solution (make certain air does not get inside). Use the hypodermic needle to guide some drops into the inlet tube of the needle from the 1mL syringe. Aim it to the side of the inlet tube to assist in preventing air bubbles from entering. If the inlet tube is filled, attach the 1mL syringe so that no air enters. Gently push the syringe so you can see that the fluid goes slowly into the inlet tube. Use plenty of light to see the front end of the fluid. If air is trapped inside, let in enough fluid to push out the bubble. If no bubble exists in the tube, take it down and fill up the Hamilton syringe.

NOTE: To prevent air from entering the Hamilton syringe, it is recommended to practice before performing the actual experiment by drawing solution up slowly, then quickly pushing back the knob of the syringe with the index finger. Push fast, but not too long. This will dispel the air. Then repeat the process until all air is eliminated. Keep the needle of the Hamilton syringe in the solution at all times during filling.

Step 3: After the Hamilton syringe is filled up, put it into the dispenser. Release some of the drug solution until a drop forms at the end of the needle. Take the filled tube/needle/1 mL syringe complex and attach it to the inlet of the electrode. Gently push the tube onto the needle, and ensure it is not leaking. At this time, inject the drug solution through the system. Use a microscope to verify the drug solution is coming out the other side. Dispel enough drug solution to eliminate any bubbles.



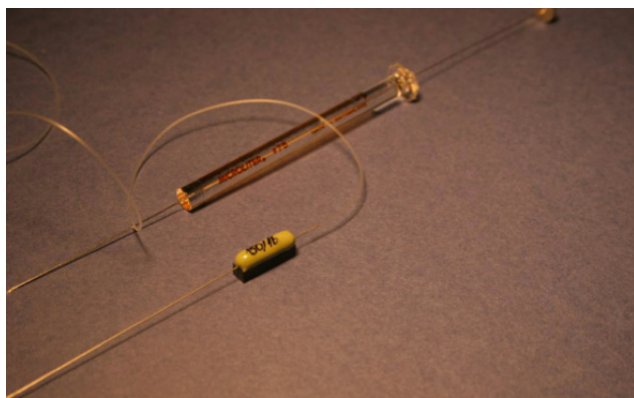
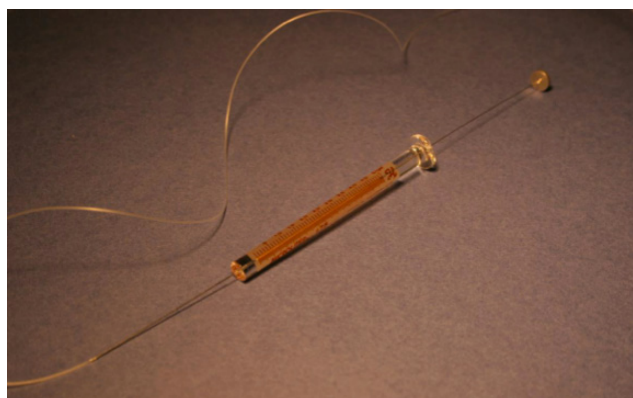
Step 4: Attach the Hamilton syringe to the tube by first removing the needle from the tube. Push gently on the 1 mL syringe and carefully remove the tube from the needle. If air remains, simply pick up the syringe with the dispenser, force out a drop of drug solution and attach the free end of the tube to the syringe. Set up is complete.

Step 5: Inject some drug solution to verify it works properly. Before implanting the probe, dip the V-Probe with the capillary into distilled water and blot up with a wet cotton tip applicator (ear cleaner) to minimize contamination.

NOTE: A typical injection volume is 0.5 - 1.0mL. If the V-Probe is in the brain, you may see from the recordings that the injection was successful. After a good injection, there is plenty of saturation or DC shift around the injection site.

Cleaning Instructions (Steps 6 - 10):

Step 6: Upon completion of the recording(s) and injection(s) with the V-Probe, gently pull the probe out of the brain and dispense the remaining drug solution from the Hamilton syringe.



Step 7: Fill the second prepared tube/needle and 5 - 10mL syringe with a combination of distilled and filtered water for use with cleaning.

Step 8: Remove the inlet tube from the electrode and attach the cleaning tube to it.

Step 9: Run a generous amount of water through the tube, and be certain the water filters out of the tube.

Step 10: Remove the tube from the electrode and force air from the third prepared tube/needle set until only air remains.

About Plexon Inc

Plexon is a pioneer and leading innovator of custom, high-performance data acquisition, behavior and analysis solutions specifically designed for scientific research. We collaborate with and supply thousands of customers including the most prestigious neuroscience laboratories around the globe driving new frontiers in areas including basic science, brain-machine interfaces (BMI), neurodegenerative diseases, addictive behaviors and neuroprosthetics. Plexon offers integrated solutions for in vivo neurophysiology, optogenetics, and behavioral research – backed by its industry-leading commitment to quality and customer support. For more information, please visit www.plexon.com.

Sales Support

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Technical Support

If after reviewing this document, you would still like to access Plexon's Technical Support, we are available via several communication channels. You are invited to reach us through email or on the phone as follows:

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